Library Management System

C language

Start

Team Members

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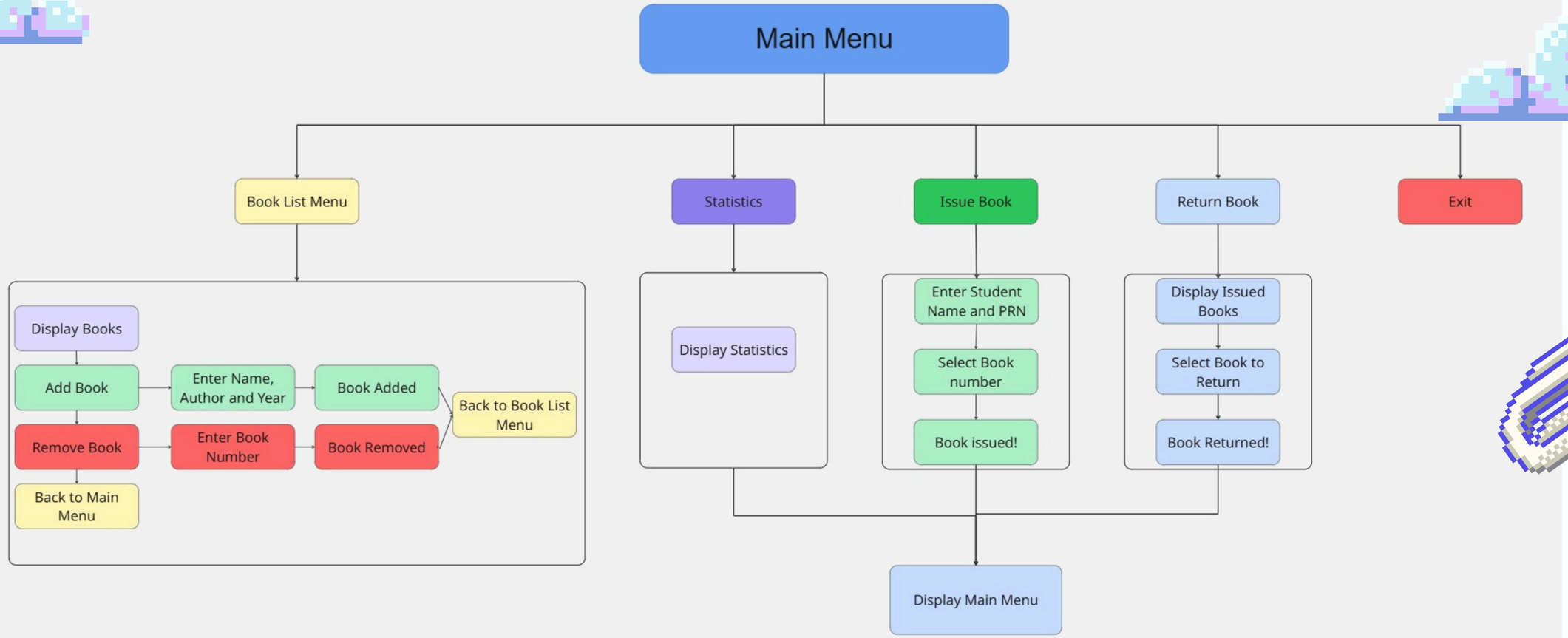
About Project

- We have made a menu-driven Library Management System using C programming language.
- It allows to manage book records by adding, removing, issuing, and returning books.
- The system keeps track of each book's availability status and records student PRN details.
- It also provides statistical information such as total books, issued books, and available books, making the library operations organized and efficient.

Concepts used

Concept	Meaning
Divide program into smaller reusable blocks	Divide program into smaller reusable blocks
Switch Case	Handle menu-based user choices
Do-While Loop	Repeat menu until user exits
Arrays	Store multiple book data
For Loop	Iterate through book records

Working of code using chart



Key Features

```
1.Harry Potter -- JK Rowling (1997) - Issued

2.Atomic Habits -- James Clear (2018) - Available

3.The Song of Ice and Fire -- George RR Martin (1996) - Available
```

```
Enter Student Name: Ritesh
Enter PRN: 1262252022
1. Harry Potter - Available
2. Atomic Habits - Available
3. The Song of Ice and Fire - Available
Select book number to issue: 1
Book issued successfully!
```

```
Total Books: 3
Issued Books: 1
Available Books: 2
```

```
--- BOOK LIST MENU ---
1. Display Books
2. Add Book
3. Remove Book
4. Back to Main Menu
-----------------------
Enter choice:
```

1. **Menu-driven interface**
2. **Add and remove book records**
3. **Issue and return books using PRN**
4. **Tracks book availability status**
5. **Displays library statistics**

Data Handling

```
char library[1000][5][50];
```

Book information is stored using a multi-dimensional character array named **library**.

```
library[i][0],/*no*/
library[i][1],/*name*/
library[i][2],/*author*/
library[i][3]);/*year*/
```

Each book record contains the book name, author, year, availability status, and student PRN.

String functions such as **strcpy()** and **strcmp()** are used to copy and compare text data.

The array structure allows efficient storage and easy updating of book records.

Thankyou!